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Topic: Large-scale Multimodal Product Classification using Autogluon (Automated ML) on high-dimensional Amazon products dataset

Abstract

Accurate product categorization is a fundamental challenge in e-commerce due to high-dimensional taxonomies and noisy metadata. This project develops a multimodal deep learning pipeline to classify 90K+ product listings into 480 unique categories using the Amazon Berkeley Objects (ABO) dataset. Using the AutoGluon's multimodal capabilities, the model fuse product image with textual metadata. It addresses the long-tail distribution of products with stratified sampling. The model achieved test accuracy of 89.4% with modest training, demonstrating robust generalization. Error analysis further demonstrated that some of the misclassification was due to semantically overlapping categories (Grocery and Nuts, Shoes and Boots), suggesting the label ambiguity.

Dataset

Amazon Berkeley Objects (ABO) - https://amazon-berkeley-objects.s3.amazonaws.com/index.html

- Creative Commons Attribution 4.0 International Public License (CC BY 4.0)
- Available for public download on Amazon S3

Technologies

- Development platform: conda venv (virtual environment), WSL, NVIDIA GPU (GeForce RTX 5070)
- Open-source libraries, frameworks: AutoGluon 1.5, PyTorch 2.9.1+cu128, pandas, sklearn

Overview of steps

1. **Data Engineering and Exploration**
    - Download the data from Amazon S3 using Bash command
        - Images: abo-images-small.tar (3 GB)
        - Labels: abo-listings.tar — Product listings and metadata (83 Mb)
    - Data quality checks and cleanup
    - Streamline data and create master data-frame using Pandas functions
    - Train/Test split (80/20) and Stratify – Product Type
2. **Autogluon Model**
    - Multimodal Prediction for Product Type – Fusing Image, Item Name, Brand
    - Model trains 73,510 items for 480 unique product types
3. **Model Evaluation and Analysis**
    - Final model gave 89.40% accuracy on the test dataset

Results

The final model gave 89.32% validation and 89.40% test accuracy on the ABO dataset. The models show strong generalization and bring forth semantically overlapping category issues in the dataset.